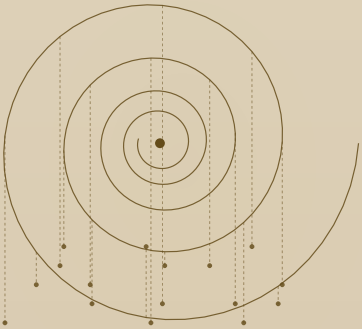
# Paper Panics



Twelve Financial Manias,
Runs, and Liquidity Crashes

ECONOMICS & FINANCIAL HISTORY

# Paper Panics

## Twelve Financial Manias, Runs, and Liquidity Crashes

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# Paper Panics

*The Hydraulic Plumbing of Financial Runs*

Financial markets look like crowds of people arguing over the future. During a panic, however, they behave like a closed hydraulic pipe under sudden pressure. In calm weather, paper claims circulate smoothly as substitutes for scarce cash. Banknotes, broker call loans, futures contracts, and derivative hedges allow thousands of participants to hold claims on the same underlying pool of coin at once. That everyday illusion of liquidity rests on a fragile statistical truce. Market plumbing assumes that claimants will never demand physical settlement at the same hour. It also assumes that falling prices will attract new buyers rather than force existing holders to sell.

A financial panic begins when a small shift in contract design, leverage, or collateral rules breaks that truce. The sign of the feedback loop suddenly reverses. When a forward obligation turns into a penalty-free call option, rational traders bid prices tenfold because they own the upside while leaving the downside behind. Suppose a bank funds twenty-year railway hulls with overnight call money, or an automated hedging algorithm sells shares whenever the index drops. Under those rules, even a minor price dip triggers mechanical liquidation. Each forced sale lowers the market price of collateral. Lower collateral prices trigger fresh margin calls across neighboring balance sheets and drain the remaining pool of buyers.

Understanding a panic requires looking past the human drama on the trading floor. One must inspect the balance-sheet ledger and the contract clause underneath. Across four centuries of paper instruments, liquidity never vanishes by mystery. It disappears when the rules of settlement force every participant toward the same narrow exit door at once.

CHAPTER 1

# The Winter Option in Haarlem

*Why Turning Tulip Contracts into Penalty-Free Options Sparked the 1637 Frenzy*

A quarto pamphlet printed in Haarlem in 1637 by the bookseller Adriaen Roman preserved a tavern dialogue between two weavers named Waermondt and Gaergoedt. Inside his ledger, Gaergoedt recorded tulip trades measured not by the flower or the sack, but by the *aas*. That unit was a Dutch goldsmith's weight equal to one-twentieth of a gram. Because tulip bulbs remain buried in frozen soil from September until June, winter traders in Haarlem, Alkmaar, and Amsterdam could not hand over physical bulbs. Instead, they met inside smoky tavern rooms in informal trading clubs called colleges. There they drank wine paid for by small transaction fees and signed notarized forward contracts promising delivery the following summer. In Roman's pamphlet, Gaergoedt boasted of the winter trade that a speculator could buy and sell millions of guilders in tulips without holding a single bulb or a single coin in his purse.

For two years after professional speculators entered the bulb trade in 1634, prices rose steadily without turning into a vertical spike. Rare broken-color varieties infected with a mosaic virus, such as *Semper Augustus* and *Viceroy*, already commanded high sums among wealthy merchant collectors. Those prized bulbs could only be propagated slowly

through vegetative offsets. Ordinary single-colored pound-goods bulbs, by contrast, traded at modest prices until the last week of November 1636. That month, setbacks in the German military campaigns cut off export demand from foreign princes. Dutch growers and tavern wholesalers were left holding forward commitments that suddenly looked underwater.

To rescue the winter trade, an assembly of Dutch florists and tavern dealers proposed a structural change to every contract signed after November 30, 1636. Instead of binding the buyer to pay the full forward price in June, every winter purchase would become a call option. When summer arrived, a buyer who found that market prices had fallen below his contract price could walk away from the deal. He only had to pay the seller a small cancellation fee called *rouwgeld*, or sorrow money, later standardized by Dutch authorities at 3.5 percent of the strike price.

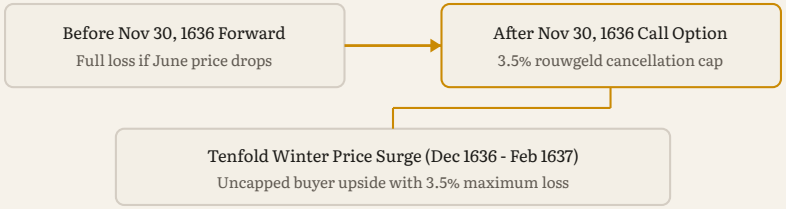



FIG. 1.1 — Contract Transformation in the November 1636 Dutch Tulip Market

That single rule change altered the payoff math of every tavern bid. A carpenter or weaver who agreed in December to pay 1,000 guilders for an *Admirael van Enckhuysen* bulb in June was no longer risking 1,000 guilders of personal ruin. If the bulb rose to 2,000 guilders, he pocketed a 1,000-guilder profit. If the market collapsed, he owed at most 35 guilders in sorrow money, and even that fee could not be collected until summer. Because downside risk was capped at 3.5 percent while upside remained unlimited, rational speculators bid aggressively for common bulbs like *Switsers* and *Witte Croonen*. Between December 1, 1636, and February 2, 1637, prices for common bulbs multiplied more than tenfold. A single pound of *Switsers* leaped from 60 guilders to 1,400 guilders.

Option manias fueled by uncollateralized promises end the moment no new buyer will accept a higher strike price. On February 3, 1637, a tavern seller in Haarlem offered a pound of bulbs at 1,250 guilders. He lowered the ask to 1,100, and then to 1,000, without receiving a single bid. Word spread by canal boat to Alkmaar and Amsterdam within forty-eight hours. On February 24, Dutch growers voted to settle all contracts signed after November 30 for a 3.5 percent fee, a compromise ratified by the Court of Holland in May. When the tulip fever broke at a Haarlem tavern auction on February 3, 1637, not a single bulb had changed hands in the winter taverns and the wider Dutch economy grew without interruption.

CHAPTER 2

# The Rue Quincampoix Printing Press

*John Law's Debt-for-Equity Swap and the Day Paper Pegged to Stock Collapsed*

Along the Rue Quincampoix, a narrow medieval lane in Paris barely fifteen feet wide and closed at both ends by iron grates, thousands of nobles and shopkeepers crushed together every morning during the autumn of 1719. Inside the packed courtyards and cafés, a humpbacked cobbler earned 150,000 livres simply by bending over. Speculators used his spine as a portable writing desk to sign share transfers. Every person in the street was trading paper issued by one man. Scottish mathematician and exiled gambler John Law had persuaded the French Regent, Philippe II, Duke of Orléans, to attempt a daring experiment. Law promised that France could erase the two-billion-livre war debt left by Louis XIV through a single grand swap of sovereign bonds into colonial equity.

Law began with sound banking mechanics. When he opened the Banque Générale in May 1716, its notes were payable on demand in coin of fixed silver weight. That guarantee made them safer than royal coins that the crown repeatedly debased. Law argued that replacing hoarded metal with circulating paper would revive frozen trade, claiming that his bank was superior to a gold mine because the rapid circulation of credit multiplied the wealth of the kingdom tenfold. Between 1717 and 1719, Law assembled the Compagnie des

Indes, known across Europe as the Mississippi Company. It absorbed the royal tobacco monopoly, the East India trade, the French mint, and the collection of all royal taxes.

To extinguish the royal debt, Law offered holders of depreciated government bonds the right to convert their state IOUs into new shares of the Mississippi Company. For the swap to succeed, the share price had to climb continuously. Law engineered the rise by turning the newly nationalized Banque Royale into a margin lender for his own stock. Anyone could buy new 500-livre shares in ten monthly installments while borrowing banknotes directly against existing shares at two percent interest. As newly printed banknotes flooded the Rue Quincampoix, Mississippi shares surged from 500 livres in May 1719 to 10,000 livres by December, minting the new French word *millionnaire*.

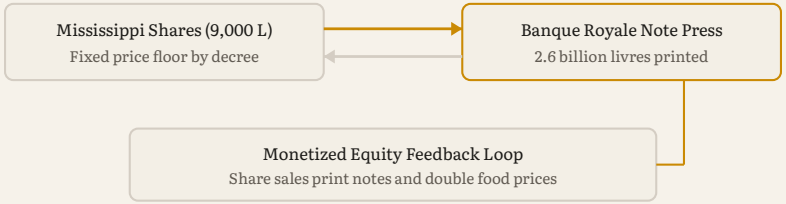



FIG. 2.1 — John Law’s 1720 Monetary Feedback Loop in Paris

At that peak, the entire company carried a paper valuation of 6.2 billion livres. Paying a four percent dividend on that sum required more silver than the entire French tax revenue

could supply. Insiders began quietly converting paper profits into gold coins, diamonds, and country estates. To stop the stock from falling in early 1720, Law committed a fatal error. He merged the Banque Royale with the Mississippi Company and pegged the share price by decree at 9,000 livres in banknotes.

Pegging a central bank's currency directly to an overvalued stock destroyed the money supply. Every time a shareholder sold one share at the official window, the Banque Royale had to print 9,000 livres of fresh banknotes. Within twelve weeks, the supply of paper money more than doubled from 1.2 billion to 2.6 billion livres. Realizing that paper notes were losing purchasing power weekly, Law issued an edict on May 21, 1720, slashing the official value of both shares and banknotes by half. Public fury forced the Regent to revoke the decree six days later, but confidence in the paper was dead. After the Prince de Conti sent three wagons to the Banque Royale in May 1720 to redeem paper notes for silver coin, French food prices doubled in six months and John Law fled Paris with only eight hundred louis d'or in his pocket.

CHAPTER 3

# The Bookseller of Exchange Alley

*How Thomas Guy Cashed Out of the South Sea Bubble While Isaac Newton Doubled Down*

A joint-stock company that possesses no profitable ships, factories, or mines can still drive its own stock price up eightfold. It only needs Parliament to grant it the right to issue new shares against rising market prices. In January 1720, the British government owed £31 million in high-interest war annuities that could not be redeemed without the bondholders' consent. Sir John Blunt, a scrivener turned director of the South Sea Company, copied John Law's Paris scheme and offered to absorb the entire national debt into South Sea shares. Under the contract approved by Parliament in April 1720, the company was authorized to issue one new share of stock for every £100 of face-value debt it retired.

That conversion rule created an accounting windfall whenever the share price climbed. If South Sea stock traded at £300, the directors only needed to hand over one-third of a share to retire £100 of sovereign debt. The remaining two-thirds of a share became surplus stock that could be sold to the public in Exchange Alley for pure cash profit. To push the stock toward £1,000, Blunt and cashier Robert Knight engineered four self-reinforcing liquidity pumps between April and August 1720:

- **Installment subscriptions**: In the four money subscriptions at £300, £400, and £1,000 per share, buyers put down only ten to twenty percent in cash. That rule turned every subscription receipt into a five-to-one leveraged call option.
- **Lending on own shares**: Between April and June, the company lent more than £11 million of newly raised subscription cash right back to shareholders at £250 to £400 per share. Borrowers used that company cash to bid up existing stock on the street.
- **Bribes in zero-money stock**: Knight entered fictitious stock options for twenty-seven members of Parliament and royal ministers, including Chancellor of the Exchequer John Aislabie. Officials collected cash price differences without paying a shilling up front.
- **The Bubble Act squeeze**: In June 1720, Blunt persuaded Parliament to outlaw rival joint-stock promotions that lacked royal charters, hoping to channel all speculative cash exclusively into South Sea shares.

One investor inside Exchange Alley understood the arithmetic of the pump from the opening bell. Veteran London bookseller Thomas Guy, the son of a Southwark lighterman, had built a fortune printing affordable English Bibles and buying discounted sailors' pay tickets. He held £54,040 of original South Sea stock acquired at roughly £120 per share. Instead of borrowing against his holdings to buy the summer subscriptions, Guy walked into Garraway's and

Jonathan's coffeehouses on April 22, 1720, when the price crossed £300. Over the next seven weeks, he sold his entire position in six disciplined tranches, disposing of his final shares on June 10 at £600.

Three miles west at the Royal Mint, Sir Isaac Newton made the exact same initial calculation. Newton sold £7,000 of South Sea stock in April and doubled his money. Then he watched his friends grow rich as the price soared past £700 in June and £950 in July. Swept into the crowd, Newton re-entered the market at the top with a vastly larger sum. According to his niece's husband, Newton later lamented that he could calculate the motions of heavenly bodies to mathematical precision, but could not calculate the madness of crowds. In August, the Bubble Act starved buyers of rival shares and forced them to dump South Sea stock to meet installment calls. When the Sword Blade Bank failed in September, liquidity vanished. While Isaac Newton re-entered the market near the summer peak and lost twenty thousand pounds, Thomas Guy used his £234,000 cash profit to construct and endow Guy's Hospital in Southwark before the stock collapsed to £124.

CHAPTER 4

# Black Friday on Lombard Street

*The Collapse of Overend Gurney and the Rule of the Lender of Last Resort*

At 3:30 p.m. on May 10, 1866, the green doors of the Corner House at 65 Lombard Street in the City of London swung shut. A clerk pinned a handwritten notice of suspension to the wood. By dawn the next day—remembered by London bankers as Black Friday—a terrified crowd packed Lombard Street so tightly that omnibuses could not pass from Mansion House to Gracechurch Street. Clerks and country bankers fought with elbows and umbrellas to reach the counters of every bill broker and joint-stock bank in the square mile. Inside the shuttered Corner House sat Overend, Gurney & Co., the largest discount house in the world. Its balance sheet exceeded the combined size of its three nearest competitors, and it had just collapsed with £11 million in unpaid liabilities.

For sixty years under its Quaker founders Samuel Gurney and Thomas Richardson, the house had operated the safest business in Victorian finance. It borrowed spare cash on overnight call from London clearing banks. It used that money to discount short-term commercial bills of exchange backed by real cotton, tea, and hardware in transit. Because sixty-day trade bills self-liquidated as goods sold, the firm earned the reputation of serving as banker to the banks. After the older generation retired in the late 1850s, younger

partners chased higher yields by quietly abandoning trade bills. They poured £4.2 million of overnight call deposits into speculative ironworks, insolvent Greek and Galway steamship fleets, and American railway contractors whose assets could not be sold in a week or a year.

When the firm converted itself into a public limited company in August 1865 to raise fresh equity from retail investors, its hidden losses were already terminal. A spring 1866 slump in railway securities and a court ruling voiding Mid-Wales Railway guarantees sparked a run by wholesale depositors. On May 10, the directors begged the Bank of England for an emergency £400,000 loan. Governor Henry Lancelot Holland sent a three-man committee to inspect Overend's ledgers. Discovering that the firm's remaining securities were rotten hulls and unmarketable paper, the Bank of England refused to lend a penny to an insolvent house.

Yet letting an insolvent giant fail created an immediate secondary crisis for every solvent firm on Lombard Street. Because country banks no longer knew which discount house held bad railway paper, everyone hoarded Bank of England notes at once. Watching the panic from his office at *The Economist*, editor Walter Bagehot distilled the crisis into his 1873 book *Lombard Street*. He laid down the classic operational protocol that separates insolvency from illiquidity:

- **Distinguish bad balance sheets from frozen markets**: Let insolvent firms like Overend, Gurney fail when their capital is gone, rather than throwing public reserves into a bottomless hole.

- **Lend freely against sound collateral**: During a general run, only the central bank's notes are trusted. To stop hoarding, the central bank must open its discount window wide to every solvent borrower holding marketable bills or government consols.
- **Charge a punitive penalty rate**: As Bagehot argued, during a panic the holders of the ultimate reserve must lend rapidly and freely to every borrower who brings sound collateral, but only at a punitive interest rate.

A high penalty rate prevents cheap speculation while rationing cash to firms that genuinely need liquidity to survive the afternoon. Once Prime Minister Gladstone suspended the gold-backing ceiling of the Bank Charter Act on the evening of May 11 and the Bank of England lent £4 million at a punitive ten percent rate, the street run halted without a single note being printed beyond the legal limit.

CHAPTER 5

# The Locked Library on Thirty-Sixth Street

*Shadow Trust Companies and How J. P. Morgan Stopped the Run of 1907*

Inside the red-silk West Room of his Renaissance marble library at East Thirty-Sixth Street and Madison Avenue in Manhattan, financier J. Pierpont Morgan sat alone at a card table playing solitaire across the first week of November 1907. In the adjoining bookrooms, two young bankers—Benjamin Strong of Bankers Trust and Willard Straight—crouched on Oriental rugs over ledgers hauled in by midnight messengers. Outside in the autumn rain, lines of depositors stretched four blocks down Fifth Avenue waiting to pull gold coin out of New York's trust companies. The United States had abolished its Second Bank in 1836 and possessed no central bank. As a result, the survival of the nation's payment grid rested entirely inside Morgan's private library.

The run of 1907 exposed a structural flaw in Gilded Age finance: the rise of an unregulated shadow banking system. Between 1897 and 1907, state-chartered New York trust companies grew nearly fourfold until their assets rivaled those of traditional national banks. Even so, they operated under completely different balance-sheet rules:

- **Regulatory reserve arbitrage**: National banks were required by law to hold twenty-five percent of their deposits in hard cash reserves. New York trust companies were re-

quired to hold only five percent in vault cash while paying higher interest rates to lure away retail depositors.

- **Call-loan maturity mismatch**: Originally chartered to manage quiet family estates, trust companies deployed their lightly reserved deposits into overnight stock-market call loans and speculative industrial syndicates. They also stood outside the protective umbrella of the New York Clearing House.
- **Contagion from a failed stock corner**: On October 16, 1907, Montana copper promoter F. Augustus Heinze and speculator Charles W. Morse tried and failed to corner the shares of United Copper. When their brokerage house collapsed, depositors learned that Knickerbocker Trust president Charles T. Barney had funded Heinze's schemes.

When the National Bank of Commerce refused to clear checks for the Knickerbocker Trust on October 21, depositors pulled $8 million from Knickerbocker's marble headquarters at Fifth Avenue and Thirty-Fourth Street in three hours. The firm shut its doors before noon. Morgan immediately dispatched Strong to audit Knickerbocker's books, found its collateral impossible to value, and let it fail. By October 23, the run leaped downtown to Oakleigh Thorne's Trust Company of America at 37 Wall Street.

Morgan ordered Strong into the vaults of the Trust Company of America overnight to price every bond and stock certificate by hand. At 12:30 p.m. on October 23, Thorne's vault cash dropped to $180,000 while a crowd clamored on Wall

Street. Strong hurried uptown and reported to Morgan that the trust company was still solvent. Morgan stubbed out his cigar and declared that the Trust Company of America was the exact line where the banking panic had to be stopped. He summoned the presidents of New York's clearinghouse banks to deliver $3 million in cash parcels within minutes. That afternoon, he also raised a $25 million call-loan pool to keep the New York Stock Exchange from shutting its doors.

Twelve days later, on November 2, a second crisis struck. The brokerage firm Moore & Schley faced bankruptcy over $25 million in loans backed by illiquid shares of the Tennessee Coal, Iron and Railroad Company. Its failure threatened to bring down the Lincoln Trust and the Trust Company of America at the start of the week. Morgan summoned over forty commercial bank and trust company presidents to his library on November 3. He placed the commercial bankers in the East Room and the trust company heads in the West Room. Then Morgan walked to the bronze front door, turned the key, and slid it into his coat pocket. He informed the trust company presidents that they had to sign a $25 million subscription to rescue their weaker peers while U.S. Steel bought Tennessee Coal. Otherwise, every bank in New York would collapse at dawn. At 4:45 a.m., the last president signed on the table. Realizing that the American financial system had survived only because one aging private banker held the keys to his library door, Congress convened the Aldrich Commission and created the Federal Reserve System in December 1913.

CHAPTER 6

# The Wheelbarrow Mark

*Thirty Paper Mills, Exploding Velocity, and the Overnight Miracle of the Rentenmark*

A one-hundred-trillion-mark banknote issued by the Weimar Reichsbank in early November 1923 carried ink on only one side of the paper and lacked a serial number. Printing both sides of the sheet or stamping sequential digits took too many seconds. In Berlin, a loaf of rye bread had risen from 250 marks in January 1923 to 200 billion marks by mid-November. Inside the Reichsbank headquarters on Jägerstraße, its veteran president, Rudolf Havenstein, viewed the crisis through an inverted lens. He treated the collapse of the German currency not as an overproduction of paper claims, but as a physical shortage of banknotes that had to be solved through industrial logistics. Addressing the Central Committee of the Reichsbank on August 17, 1923, Havenstein boasted that thirty paper mills were running at full capacity and would soon deliver two-thirds of a quadrillion marks every single day.

By October 1923, 30 paper mills, 133 printing firms, and 1,783 high-speed printing presses ran three shifts around the clock. Freight trains hauled bundles of freshly cut marks into the Ruhr valley to pay striking miners during the French occupation. Yet the harder Havenstein ran his presses, the more desperately German cities complained of a severe

shortage of circulating money. Municipalities and factories began printing their own emergency scrip (*Notgeld*) just to meet daily payrolls.

That apparent paradox—an acute shortage of money during the largest printing binge in European history—follows directly from Irving Fisher's equation of exchange. The law relates the nominal money supply $M$, the velocity of circulation $V$, the general price level $P$, and real physical output $Y$:

$$M \cdot V = P \cdot Y \Longrightarrow \frac{M}{P} = \frac{Y}{V}$$

During the early stages of German inflation between 1919 and mid-1922, prices rose roughly in proportion to the printing of new marks. Foreign speculators still bought German paper, hoping the mark would eventually recover. Once French and Belgian troops marched into the Ruhr in January 1923, the Reichsbank began discounting both government treasury bills and private commercial bills at a deeply negative real interest rate. At that moment, public expectations snapped. Everyone realized that holding a paper mark for even four hours meant losing half its value.

When a population flees its currency, the velocity of circulation $V$ explodes vertically. Factory workers demanded wages twice a day, at 11:00 a.m. and 4:00 p.m. They tossed bundles of marks out factory windows to spouses waiting below. Those spouses sprinted directly to bakeries and coal yards to exchange paper for physical calories before the noon telegraph

from New York posted the new dollar exchange rate. Because $V$ multiplied far faster than Havenstein's 1,783 presses could increase $M$, the price level $P$ outran the money supply. Consequently, the real purchasing power of the entire German money stock ($M/P$) shrank week after week.

Stopping a velocity spiral requires breaking the expectation of tomorrow's printing press. Following Havenstein's sudden death from a heart attack in November 1923, newly appointed Currency Commissioner Hjalmar Schacht executed a coordinated monetary halt between November 15 and November 20. The government introduced a new unit, the Rentenmark, backed on paper by a statutory collective mortgage on German farmland and factories. Schacht fixed the exchange rate at one Rentenmark—twenty-four American cents—for one trillion ($10^{12}$) old paper marks, pegging one U.S. dollar at 4.2 trillion marks. The real anchor was not the unredeemable land mortgage, but Schacht's iron refusal to discount another government treasury bill or bail out speculators who had shorted the currency. At the peak of hyperinflation in October 1923, workers spent their wages so fast that the entire paper currency of Germany was worth less in gold than one-tenth of the pre-war coin supply.

CHAPTER 7

# The Ten-Percent Margin Trap

*Call-Money Loans and Why Paying Down Debt Deepened the Crash of 1929*

A retail investor who walked into a Manhattan brokerage office in the summer of 1929 could purchase $100 worth of Radio Corporation of America stock by handing the clerk just $10 in cash. The remaining $90 came from the New York Stock Exchange's call-money desk, where brokers borrowed overnight funds secured by the customer's stock certificates. Call loans could be terminated on twenty-four hours' notice whenever a share price dipped toward the ten-percent equity cushion. Thus, the entire late-1920s bull market rested on a hair-trigger collateral chain. By early October 1929, total broker call loans reached $8.5 billion, exceeding the entire circulating currency of the United States. Over $6.6 billion of that money came not from regulated banks, but from corporate treasurers at companies like Standard Oil and Electric Bond and Share chasing twelve-percent overnight interest rates.

Among the most heavily leveraged buyers was Yale economist Irving Fisher, the preeminent monetary theorist in America. Fisher had also invented the visible-index rotary card file, selling his company to Remington Rand for a fortune exceeding $10 million. Convinced that industrial electrification had placed equities on a permanently high plateau, Fisher borrowed millions on margin to accumulate

additional shares right into the autumn of 1929. When prices slipped in mid-October, non-bank corporate lenders suddenly yanked their call money out of Wall Street to protect their own treasuries. Brokers had no choice but to send immediate morning telegrams demanding cash margin from their customers.

On Black Thursday, October 24, and Black Tuesday, October 29, 1929, customers who could not wire cash by 10:00 a.m. had their shares dumped unconditionally at market open. Sixteen million shares changed hands on October 29 as the ticker tape fell two and a half hours behind the floor. Fisher lost his entire $10 million fortune and his New Haven mansion. He had to borrow from his sister-in-law and Yale University just to keep a roof over his head. Searching the wreckage to understand why the economy did not bounce back after the initial liquidation, Fisher published his landmark 1933 *Econometrica* paper on the debt-deflation theory of great depressions.

Fisher proved that when an entire economy is over-indebted on short-term margin, universal liquidation triggers a mathematical trap. Suppose borrowers start with nominal debt $D_0$ at an initial price level $P_0$. Margin calls then force everyone to dump stocks, commodities, and bank assets simultaneously to pay down a fraction $\alpha$ of their nominal dollar debt. That stampede collapses the velocity of money and drives the general price level down by a fraction $\beta$. The resulting real debt burden $D_{\text{real}}$, measured in purchasing-

power units relative to the starting burden $D_0/P_0$, obeys the ratio:

$$D_{\text{real}} = \frac{D_0(1-\alpha)}{P_0(1-\beta)}$$

Whenever distress selling depresses the general price level faster than nominal debts are retired ($\beta > \alpha$), the denominator shrinks faster than the numerator. The real debt burden grows larger than before a single dollar was repaid. As Fisher explained in 1933, the more debtors pay down their nominal balances during a panic, the more they owe in real purchasing power. The collective stampede to liquidate assets defeats each individual's effort to reduce debt. Between October 1929 and March 1933, American borrowers paid off twenty percent of their nominal dollar debts, yet because forced liquidation drove the price level down by forty percent, every remaining dollar of debt weighed thirty-three percent more in real terms.

CHAPTER 8

# Silver Rule Seven

*How the Hunt Brothers Cornered Two Hundred Million Ounces and Lost to a Rule Change*

Cornering a physical commodity requires solving a two-sided equation. A buyer must accumulate enough warehouse receipts to force short sellers into a delivery squeeze while borrowing cash against the rising price of the hoard itself. In 1973, Dallas oil billionaire Nelson Bunker Hunt and his brother William Herbert Hunt—sons of wildcatter H. L. Hunt—sought a tangible hedge against paper inflation. Libya had just nationalized their oil fields, and United States law still barred private citizens from owning gold bullion. Dismissing suggestions that two private investors could dominate global bullion supplies, Bunker Hunt told reporters that anyone who believed a pair of Texans could corner the world silver market had no grasp of how vast the world actually was.

Between 1974 and 1979, the brothers set out to test that proposition on the New York Commodity Exchange (COMEX) and the Chicago Board of Trade. Instead of settling futures contracts in cash as normal speculators did, the Hunts demanded physical delivery of 5,000-ounce silver bars. They chartered Boeing 707 cargo jets with reinforced floors to fly forty million ounces of silver bullion to three private vaults outside Zurich. By mid-1979, partnering with Saudi investors

through a Bermudian shell company named International Metals Investment Company, the Hunts controlled roughly 100 million ounces of physical silver and another 100 million ounces in futures contracts. Together, those holdings represented one-third of the non-communist world's deliverable supply.

As silver vaulted from $6 an ounce in January 1979 to an intraday peak of $50.35 on January 18, 1980, the brothers used their mark-to-market paper profits as collateral. They borrowed fresh bank loans to buy still more futures contracts. Their leveraged corner collided with four sequential shocks between January and March 1980:

- **Scrap supply elasticity**: Drawn by fifty-dollar silver, millions of households queued outside coin dealers to melt Victorian tea sets, wedding flatware, and pre-1965 quarters. Refineries were flooded with unexpected physical metal while industrial users like Eastman Kodak and Tiffany & Co. protested shortages.
- **Margin hikes by shorts on the board**: Several COMEX board members held commercial short hedges that were bleeding cash on daily margin calls. Between late 1979 and January 1980, the exchange hiked initial margin requirements per 5,000-ounce contract from $1,000 to $75,000.
- **Silver Rule 7 (liquidation-only trading)**: On January 21, 1980, COMEX Chairman Michel Marks announced an emergency directive known as Silver Rule 7. It barred any trader from opening new long futures positions and

restricted all trading strictly to the liquidation of existing contracts.

- **The Volcker interest-rate squeeze**: While Silver Rule 7 outlawed every potential buyer who might have bid prices higher, Federal Reserve Chairman Paul Volcker pushed the prime lending rate to twenty percent. That hike multiplied the Hunts' monthly interest bill on their bullion loans.

With buying outlawed and selling mandatory, the price of silver had only one direction to travel. Each dollar decline below $35 triggered automatic variation margin calls against the Hunts' leveraged futures accounts. To stay afloat, they borrowed over $800 million from commercial banks and their brokers, Bache Halsey Stuart Shields. After the close on March 25, 1980, Bache demanded $135 million in immediate cash margin. Bunker Hunt informed the brokerage from Saudi Arabia that the brothers had no more cash. On Thursday, March 27—remembered on Wall Street as Silver Thursday—Bache began dumping the Hunts' silver collateral on the open floor to save itself. Once COMEX prohibited new long buyers and raised margin requirements to seventy-five thousand dollars per contract, silver plunged from fifty dollars to ten dollars and eighty cents, forcing the Hunts to pledge their Placid Oil empire for a $1.1 billion rescue loan.

CHAPTER 9

# The Doomsday Machine of Black Monday

*How Automated Portfolio Insurance Turned a Dip into a Twenty-Two Percent Crash*

An individual pension fund can protect its stock portfolio against a market crash without buying a single put option if it continuously adjusts its cash balance according to a mathematical formula. When sixty billion dollars of institutional capital execute that same formula at the same hour, however, the hedge creates the crash it was built to prevent. In 1976, after watching endowments flee the stock market during the 1973–1974 downturn, Berkeley finance professor Hayne Leland woke up with an insight derived from the 1973 Black-Scholes option-pricing equation. Together with colleague Mark Rubinstein and marketing executive John O'Brien, Leland founded Leland O'Brien Rubinstein Associates (LOR) in 1981 to sell a computerized product they called portfolio insurance.

Long-term protective put options did not trade in size for massive corporate pension funds in the early 1980s. Leland and Rubinstein showed that a computer could manufacture a synthetic protective put dynamically by holding a shifting blend of S&P 500 stocks and Treasury bills. Under Black-Scholes replication, two numbers govern the computer's trades: the synthetic put's hedge ratio ($\Delta_{\text{put}}$, the fraction of the portfolio that must be sold as prices decline) and its

curvature sensitivity ($\Gamma$, the rate at which required selling accelerates for every point drop in the index $S$ at volatility $\sigma$ and horizon $T$). Both follow directly from the cumulative normal distribution $N(d_1)$:

$$\Delta_{\text{put}} = N(d_1) - 1, \qquad \Gamma = \frac{\partial \Delta}{\partial S} = \frac{N'(d_1)}{S\sigma\sqrt{T}} > 0$$

Because gamma is strictly positive, replicating a protective floor requires a trend-following feedback rule. Whenever the stock market rises, the computer buys more equity exposure. Whenever the stock market falls, the computer must mechanically sell stocks and move into cash so the portfolio cannot breach its insured floor. To avoid paying commissions on five hundred individual equities, LOR and its competitors executed these dynamic sales by dumping S&P 500 index futures contracts on the Chicago Mercantile Exchange.

By October 1987, institutional funds had placed more than $60 billion under automated portfolio-insurance rules. None of the Black-Scholes models accounted for what happens when the hedger becomes the entire market. As Rubinstein later acknowledged, portfolio insurance worked smoothly only so long as a single investor used it, much like a theater fire exit that functions well only when a single patron runs for the door. During the week of October 14–16, 1987, rising interest rates, a falling dollar, and Congressional tax proposals pushed the Dow Jones Industrial Average down

ten percent. That slide left portfolio-insurance computers with a huge backlog of unexecuted sell orders over the weekend.

When the bell rang on Monday, October 19, 1987, LOR and rival insurers flooded the Chicago S&P 500 futures pit with mechanical sell orders. Futures immediately dropped to a steep discount below the cash stock index in New York. Index arbitrageurs stepped in to profit from the spread. They bought cheap Chicago futures and simultaneously transmitted automated Designated Order Turnaround (DOT) orders to sell the underlying stocks on the New York Stock Exchange. That arbitrage transmitted the futures avalanche directly onto NYSE floor specialists, whose capital was exhausted before noon. As New York stock prices plunged, the lower index values fed back into the portfolio-insurance computers. Their positive-gamma equations calculated that $\Delta_{\text{put}}$ had grown even more negative and commanded another wave of futures sales regardless of price. By the 4:00 p.m. close, the Dow had fallen 508 points, losing 22.6 percent in a single session. Because dynamic delta-hedging requires selling shares whenever prices fall, the sixty billion dollars of insured pension portfolios acted as a synchronized mechanical seller that overwhelmed every human buyer on the floor in five hours.

CHAPTER 10

# The Broken Pound on Black Wednesday

*George Soros, the Impossible Trinity, and the Ten-Billion-Dollar Wager Against the Bank of England*

Late in the afternoon of Tuesday, September 15, 1992, inside the Manhattan offices of the Quantum Fund on Seventh Avenue, chief portfolio manager Stanley Druckenmiller walked into George Soros's office to lay out a currency trade. Britain had joined the European Exchange Rate Mechanism (ERM) two years earlier at a central peg of 2.95 Deutsche Marks to the pound. That afternoon, German Bundesbank president Helmut Schlesinger had hinted to reporters that European currencies needed a broader realignment. Druckenmiller explained why the British pound could not hold its floor and proposed building a $1.5 billion short position. Soros listened to the asymmetric math of the trade and frowned at the modest size, asking why they were stopping at two billion dollars if the thesis was sound and urging Druckenmiller to go for the jugular.

Overnight between September 15 and the London open on Wednesday, September 16, 1992, Quantum Fund traders borrowed and sold more than $10 billion worth of British pounds for Deutsche Marks. What made the trade nearly risk-free for Soros was a structural contradiction in international macroeconomics known as the Mundell-Fleming Impossible Trinity. A sovereign nation can choose at most

two of three policy goals simultaneously: free cross-border capital flows, a fixed exchange rate, and an independent domestic interest-rate policy tailored to its own business cycle.

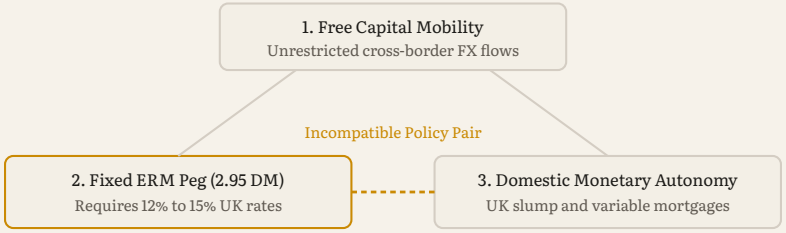



FIG. 10.1 — The Impossible Trinity in the September 1992 British Sterling Crisis

In the autumn of 1992, Germany and Britain sat at opposite ends of the economic cycle. Following the reunification of East and West Germany in 1990, Bonn spent heavily on eastern reconstruction. That spending stoked inflation and prompted the Bundesbank to raise German interest rates to 8.75 percent. Britain, by contrast, was mired in a deep housing and manufacturing recession. Unemployment had climbed past 2.8 million, and ninety percent of British homeowners held floating-rate mortgages tied directly to the Bank of England's base rate.

To keep the pound inside its narrow ERM band above 2.778 Deutsche Marks while capital flowed freely across Europe, Britain had only two tools. It could buy pounds on the open market using finite foreign-exchange reserves, or it

could match and exceed German interest rates. Both tools contained an obvious breaking point. If the Bank of England defended the peg at 2.778, Soros and Druckenmiller knew the pound would never be revalued upward during a British recession. Their upside risk on a short position was capped near zero, whereas a devaluation promised an instant ten-to-fifteen-percent windfall.

When London markets opened at 7:30 a.m. on Black Wednesday, September 16, Bank of England dealers at Threadneedle Street faced a wall of sterling sell orders from Quantum, pension funds, and corporate treasurers. By 9:00 a.m., the Bank was buying two billion pounds an hour just to hold the 2.778 floor. At 10:30 a.m., Prime Minister John Major and Chancellor of the Exchequer Norman Lamont authorized an emergency interest-rate hike from 10 percent to 12 percent. Currency traders barely blinked. Everyone knew that a twelve-percent mortgage rate would bankrupt British households and deepen the slump. At 2:15 p.m., a desperate Lamont announced a second rate hike to 15 percent effective the next morning. Selling accelerated immediately because every desk on the street realized the political bluff had failed. At 7:40 p.m., Lamont stepped outside 11 Downing Street to announce Britain's suspension from the ERM, netting Soros's fund over $1 billion in profit. While the Bank of England burned through twenty-seven billion dollars of reserves and hiked interest rates to fifteen percent in a single morning, Britain's exit from the peg lowered borrowing

costs and launched sixteen consecutive years of economic expansion.

CHAPTER 11

# Account Five Eights in Singapore

*Nick Leeson, Short Straddles, and the Earthquake That Brought Down Barings Bank*

Cleaning staff in Room 1106 of the Regent Hotel in Kuala Lumpur found an abandoned keycard on February 23, 1995. On the desk lay a brief handwritten apology note. Its author, Nick Leeson, the general manager and head floor trader for Barings Futures Singapore, was boarding a flight to Borneo and then Frankfurt while auditors in Singapore pried open his locked filing cabinets. Inside Barings' computer system sat a client error account numbered 88888, chosen because eight is a lucky digit in Chinese numerology. That single file concealed £827 million ($1.3 billion) in unhedged derivative losses. Within forty-eight hours, it wiped out 233-year-old Barings Bank, the oldest merchant bank in London and personal banker to Queen Elizabeth II.

Leeson had arrived on the floor of the Singapore International Monetary Exchange (SIMEX) in 1992 with a narrow, low-risk mandate. His job was to arbitrage tiny price discrepancies in Nikkei 225 stock-index futures between the open-outcry pit in Singapore and the electronic exchange in Osaka. Because Barings failed to separate trading from back-office settlement, Leeson served simultaneously as chief floor trader and head of accounting operations. As Leeson later recounted in his memoir of the collapse, executives at

the London end of Barings projected such omniscient confidence that nobody dared ask a basic question for fear of looking foolish in front of colleagues.

When one of his clerks accidentally bought twenty Nikkei futures contracts instead of selling them on July 17, 1992, creating a £20,000 loss, Leeson hid the trade inside error account 88888 rather than reporting it to London. Discovering that he could book losing trades to 88888 while crediting offsetting fictitious gains to his public trading accounts, Leeson began doubling down on directional bets whenever the Tokyo market moved against him. By the end of 1993, hidden losses inside 88888 reached £23 million. By December 1994, they hit £208 million, even as London executives paid Leeson a £130,000 bonus and praised him for generating a fifth of Barings' global profits.

Holding thousands of losing futures contracts on SIMEX required wiring hard cash to the exchange clearinghouse every morning to meet variation margin calls. To raise upfront cash without asking London for every yen, Leeson turned to selling Nikkei 225 option straddles. A short straddle consists of simultaneously selling a call option and a put option at the exact same strike price $K$ and expiration date $T$, collecting the combined upfront premiums $P_{\text{call}} + P_{\text{put}}$. At expiration, the seller's net payoff $\Pi(S_T)$ as a function of the final index level $S_T$ is:

$$\Pi(S_T) = P_{\text{call}} + P_{\text{put}} - \mid S_T - K \mid$$

Selling straddles is a pure bet against volatility. If the Nikkei index stays pinned near the strike price $K$ ($18,500$ to $19,500$), the trader keeps the upfront cash premiums to pay his futures margin calls. If the index moves sharply in either direction, however, losses grow linearly without bound. By January 1995, Leeson had sold roughly 37,000 short straddles—74,000 option contracts—turning Barings into the primary volatility insurer of the entire Japanese stock market.

At 5:46 a.m. on Tuesday, January 17, 1995, a magnitude 6.9 earthquake struck Kobe, Japan, destroying ports, highways, and over one hundred thousand buildings. As the Nikkei 225 dropped more than a thousand points over the week of January 23, Leeson suffered a double hemorrhage. His short puts exploded in liability under $-\mid S_T - K \mid$. In a desperate attempt to push the entire Japanese stock market back up into his $19,000$ straddle cone, he bought 61,039 March Nikkei futures contracts—half the open interest on SIMEX—while tricking London treasurers into wiring £500 million to cover bogus client loans. When the Kobe earthquake knocked the Nikkei index down past eighteen thousand points, Leeson's unhedged futures and option losses reached £827 million—twice the entire equity capital of 233-year-old Barings Bank, which was sold to ING for one pound.

CHAPTER 12

# The Hot-Potato Algorithm

*How a Volume-Pegged Sell Order Erased a Trillion Dollars in Thirty-Six Minutes*

A single line of execution parameters recorded on the joint CFTC-SEC audit trail from Thursday, May 6, 2010, captures how modern electronic liquidity can evaporate in minutes. At 2:32:00 p.m. Eastern Time, European sovereign debt fears over Greece were already weighing on the market. In Overland Park, Kansas, a mutual fund complex named Waddell & Reed Financial initiated an automated program to hedge its equity portfolio. It entered an order to sell 75,000 June 2010 E-Mini S&P 500 futures contracts worth roughly $4.1 billion on the Chicago Mercantile Exchange's Globex electronic platform. Previously, executing a $4 billion hedge took more than five hours of careful slicing. That afternoon, the firm selected an off-the-shelf broker algorithm programmed with a single mechanical rule: as the official regulatory report later documented, the software was instructed to target nine percent of the trading volume calculated over the previous minute without regard to price or time.

Targeting a fixed percentage of rolling one-minute trading volume without a minimum price floor or a maximum execution speed created a closed positive feedback loop the moment it met high-frequency market makers. Between 2:32 p.m. and 3:08 p.m., the interaction of those two automated

systems erased nearly one trillion dollars of market value across four distinct mechanical phases:

1. **Initial liquidity absorption (2:32 p.m. to 2:41 p.m.)**: During the first nine minutes, the Waddell & Reed sell algorithm dumped 35,000 E-Mini contracts into the Globex order book. High-frequency trading (HFT) firms, traditional buyers, and cross-market arbitrageurs absorbed those initial contracts. The selling pushed S&P 500 futures down three percent while exhausting the top layers of resting buy orders.
2. **Inventory saturation and hot-potato churn (2:41 p.m. to 2:45:27 p.m.)**: Once high-frequency market makers hit their strict intraday position limits of 2,000 to 3,000 contracts, their algorithms immediately flipped from net buyers to aggressive sellers to dump inventory before prices fell further. They sold contracts to other HFT algorithms, which instantly resold them back. The Waddell & Reed algorithm measured total traded volume over the prior sixty seconds without knowing that HFTs were merely passing the same contracts back and forth. It interpreted the churn as a massive surge in market liquidity and ramped up its own selling speed to match nine percent of the artificial volume spike.
3. **Cross-market contagion and stub quotes (2:45 p.m. to 2:47 p.m.)**: Arbitrage computers buying cheap E-Mini futures in Chicago simultaneously fired rapid-fire sell orders into the S&P 500 ETF (SPY) and individual equities in New York. Sensing abnormal velocity, automated eq-

uity market makers switched off their quoting engines to avoid toxic order flow. With real bids withdrawn, sweeping market orders hit placeholder "stub quotes" posted at a single penny or $100,000 to satisfy nominal exchange rules. Blue-chip shares of Accenture and Procter & Gamble plunged from $40 to $0.01 in seconds while the Dow Jones Industrial Average dropped 998.5 points.

4. **The five-second circuit breaker (2:45:28 p.m. to 3:08 p.m.)**: At 2:45:28 p.m., after the E-Mini had plunged five percent in four minutes and finished digesting all 75,000 Waddell & Reed contracts in just twenty minutes, CME Globex's automated Stop Logic Functionality triggered a five-second trading pause.

Those five seconds of forced silence broke the volume feedback loop and reset the rolling one-minute volume counter to zero. Human and algorithmic desks reposted real bids when trading resumed at 2:45:33 p.m. During a single fourteen-second window at 2:45 p.m., high-frequency trading firms passed twenty-seven thousand futures contracts back and forth while changing their net inventory by fewer than two hundred contracts.